

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250278973

Kind Code

A1

Publication Date

September 04, 2025

Inventor(s)

BYTNAR; Michael R. et al.

PERSISTENT DEVICE RELATIONSHIPS IN WAGERING GAME SYSTEMS

Abstract

Example systems, methods, and media facilitate establishing secure communication sessions between a wagering game machine and a wireless device. Upon authentication of a player account at a first wagering game machine and detection of a trigger condition for interaction with an associated wireless device, derived authentication data or pairing information is automatically obtained. This data/information is based on secret/pairing information (e.g., device ID, PIN) previously stored from an initial pairing of the wireless device at an additional wagering game machine and linked to the player account. The first wagering game machine uses the obtained data/information, which may be received from the wireless device or a network configuration/storage device, to automatically establish a secure session with the wireless device, potentially via a cryptographic authentication process.

Inventors: BYTNAR; Michael R. (Naperville, IL), GAGNER; Mark B. (West Chicago, IL), LOOSE; Timothy C. (Chicago, IL), PACEY; Larry J. (Calabasas, CA), SCHULHOF; Scott H. (Chicago, IL)

Applicant: LNW Gaming, Inc. (Las Vegas, NV)

Family ID: 41016715

Appl. No.: 19/212197

Filed: May 19, 2025

Related U.S. Application Data

parent US continuation 17738795 20220506 parent-grant-document US 12315325 child US 19212197

parent US continuation 16549949 20190823 parent-grant-document US 11335155 child US 17738795

Publication Classification

Int. Cl.: **G07F17/32** (20060101); **H04L67/131** (20220101); **H04W12/06** (20210101);
H04W12/50 (20210101)

U.S. Cl.:

CPC **G07F17/32** (20130101); **G07F17/3218** (20130101); **G07F17/3223** (20130101);
H04L67/131 (20220501); **H04W12/06** (20130101); **H04W12/50** (20210101);

Background/Summary

RELATED APPLICATIONS [0001] This application is a continuation of U.S. patent application Ser. No. 17/738,795, filed May 6, 2022, which is a continuation of U.S. patent application Ser. No. 16/549,949, filed Aug. 23, 2019 (now U.S. Pat. No. 11,335,155), which is a continuation of U.S. patent application Ser. No. 12/918,066, filed Aug. 18, 2010 (now U.S. Pat. No. 10,395,465), which is the National Stage of International Application No. PCT/US2009/035355, filed Feb. 26, 2009, which claims the priority benefit of Provisional U.S. Patent Application No. 61/031,923, filed Feb. 27, 2008. Each of the foregoing applications and patents is incorporated by reference herein in its entirety.

LIMITED COPYRIGHT WAIVER

[0002] A portion of the disclosure of this patent document contains material which is subject to copyright protection. The copyright owner has no objection to the facsimile reproduction by anyone of the patent disclosure, as it appears in the Patent and Trademark Office patent files or records, but otherwise reserves all copyright rights whatsoever. Copyright 2025, LNW Gaming, Inc.

FIELD

[0003] Embodiments of the inventive subject matter relate generally to wagering game systems, and more particularly to methods and systems for establishing secure communication sessions between wagering game machines and wireless devices.

BACKGROUND

[0004] Wagering game machines, such as slot machines, video poker machines, and the like, have been a cornerstone of the gaming industry. The popularity of such machines often depends on the likelihood of winning and the intrinsic entertainment value. Operators of gaming establishments strive to employ machines with engaging features and enhancements to attract frequent play and thereby increase profitability. Consequently, there is a continuing need for wagering game machine manufacturers to develop new games and gaming enhancements that improve player experience, streamline interactions, and maintain security. One area for enhancement involves the interaction between players' personal wireless devices and wagering game machines, particularly in establishing and maintaining secure communication for various game-related functions after an initial trust relationship has been formed.

SUMMARY

[0005] According to aspects of the present disclosure, a method, system, etc. is provided for establishing secure sessions. The approach entails authenticating a player account using, at least in part, a player identifier received at a wagering game machine (identified as a first wagering game machine in method claims). The operations also entail detecting a trigger condition indicating potential interaction with a wireless device associated with the authenticated player account. Further, operations include, in response to detecting the trigger condition, automatically receiving,

without user input, derived authentication data directly from the wireless device or from a configuration device accessible via a network via the communication interface. This derived authentication data is based upon secret information, including a wireless device identifier and a personal identification number (PIN) of the wireless device, the secret information having been previously stored in association with the player account information by an additional processor of an additional wagering game machine (identified as a second wagering game machine in method claims) communicatively connected to a network, in response to a previous pairing by the wireless device with that additional wagering game machine. The method and system operations also feature automatically establishing, by the processor, a secure communication session between the wagering game machine and the wireless device based, at least in part, on the received derived authentication data.

[0006] Additionally, aspects of the disclosure relate to a method, system, etc. that performs operations that include receiving a player identifier associated with an authenticated player account. The operations further entail, in response to receiving the player identifier, determining whether pairing information associated with the player identifier for a specific wireless device is stored on a centralized storage device accessible via a communication network; this pairing information was previously stored on the centralized storage device based on a previous pairing or registration involving the specific wireless device, and the pairing information includes one or more security elements based upon the specific wireless device. The operations also include, in response to determining that the pairing information is stored on the centralized storage device, retrieving, via the communication network, the stored pairing information from the centralized storage device. The wagering game machine is also caused to automatically establish a secure communication session with the specific wireless device based, at least in part, on the retrieved pairing information, wherein said automatically establishing the secure communication session includes performing a cryptographic authentication process with the specific wireless device utilizing the retrieved pairing information.

Description

BRIEF DESCRIPTION OF THE FIGURES

[0007] Example embodiments of the present disclosure are illustrated in the Figures of the accompanying drawings in which:

[0008] FIG. 1 is a conceptual diagram illustrating how a wagering game system can persistently store and reuse PAN device PINs, according to some embodiments of the invention;

[0009] FIG. 2 is a block diagram illustrating a wagering game machine architecture, according to example embodiments of the invention;

[0010] FIG. 3 is a block diagram illustrating a wagering game network 200, according to example embodiments of the invention;

[0011] FIG. 4 is a flow diagram illustrating operations for pairing a wagering game machine with a PAN device, according to some embodiments of the invention;

[0012] FIG. 5 is a flow diagram illustrating operations for pairing a PAN device to a wagering game machine, according to some embodiments of the invention;

[0013] FIG. 6 is a flow diagram illustrating operations for using PAN devices to present community wagering game results, according to some embodiments of the invention; and

[0014] FIG. 7 is a perspective view of a wagering game machine, according to example embodiments of the invention.

DESCRIPTION OF THE EMBODIMENTS

[0015] This description of the embodiments is divided into five sections. The first section provides an introduction to embodiments of the invention, while the second section describes example

wagering game machine architectures. The third section describes example operations performed by some embodiments and the fourth section describes example wagering game machines in more detail. The fifth section presents some general comments.

INTRODUCTION

[0016] This section provides an introduction to some embodiments of the invention.

[0017] Wagering game systems can include personal area networking (PAN) technologies (e.g., Bluetooth™, IrDA®, ZigBee, Ultra-wideband, etc.) that enhance gaming experiences. Some PAN technologies allow players to provide input and receive output using wireless devices. For example, PAN devices can enable players to receive audio via wireless headphones, provide input via wireless input devices (e.g., Nintendo Wii), receive content on personal digital assistants or smartphones, provide speech input via wireless microphone, etc.

[0018] To establish secure communications, PAN devices and wagering game machines may perform operations for “pairing.” Pairing is a process in which a wagering game machine authenticates a PAN device using a secret shared between the devices. In some instances, a player enters a secret, such as a personal identification number (PIN) or other authentication code including any suitable indicia such as letters and numbers, into a wagering game machine and a PAN device. The wagering game machine and PAN device can use the PIN to create encryption keys which they can use for authentication and establishing encrypted communications. After a player initially provides a PAN device's PIN, some embodiments enable the player to use the PAN device with any other wagering game network device (e.g., other wagering game machines) without the player again providing the PAN device's PIN. FIG. 1 describes these concepts in more detail.

[0019] FIG. 1 is a conceptual diagram illustrating how a wagering game system can persistently store and reuse PAN device PINs, according to some embodiments of the invention. In FIG. 1, six stages show how a wagering game system **100** stores and reuses PAN device PINs. In FIG. 1, the wagering game system **100** includes two wagering game machines **102** & **118** and an account server **116**.

[0020] During stage one, a player initiates a wagering game session on a wagering game machine **102** by swiping a player card **104** through a card reader **106**. Swiping the player card **104** provides the player's player identifier to the wagering game machine **102**. The wagering game machine may also ask the player to enter a password (or other secret), so it can authenticate the player. During stage two, the wagering game machine **102** looks for PAN devices with which it can pair. After recognizing the PAN device **102**, the wagering game machine **102** prompts the player **110** to enter a PIN associated with the PAN device **102**. As shown, the PAN device **112** can be wireless headphones or any other suitable device, such as a PDA, media player, remote control, etc.

[0021] During stage three, the wagering game machine **102** establishes communications with the PAN device **112** and uses the PIN to authenticate the PAN device **112**. After authentication, the device **112** and machine **102** can securely communicate using encrypted communications. The wagering game machine **102** can present content and game results on its own components, while also sending content and game results (e.g., a game's audio content) for presentation on the PAN device **112**.

[0022] During stage four, the wagering game machine **102** transmits the player's identifier, an identifier for the PAN device **112**, and the PAN device's PIN to the account server **116** (see **114**). The account server **116** can store the PAN device identifier and PIN with the player's account information (e.g., account balance, player preferences, other PAN device PINs, etc.). The PAN device's PIN can be used later when the PAN device **112** pairs with other wagering game machines, as described below.

[0023] During stage five, the player initiates a wagering game session with another wagering game machine **118**. To initiate the session, the player swipes the player card **104** through the machine's card reader **120**, providing the player's player identifier to the wagering game machine **118**. After

receiving the player identifier, the wagering game machine **118** asks the account server **116** for any PAN device identifiers and PAN device PINs associated with the player identifier (see **124**). During stage six, the wagering game machine **102** looks for any PAN devices with which it can pair. Using the PAN device identifier and PIN received from the account server **116**, the wagering game machine **118** authenticates the PAN device **112**. In turn, the wagering game machine **118** sends wagering game content to the PAN device **112**.

[0024] Although some of the above-described embodiments pair using PINs, other embodiments use other techniques. For example, some embodiments can require that secret information (e.g., a private key) be transmitted to one or both devices via a secure means, such as over an encrypted wireless connection or a direct and secure wired connection. In some instances, a separate “configuration device” can provide the secret information to both the PAN device and wagering game machine. Alternatively, a wagering game machine or PAN device can provide the secret information to the other. After the initial transmission of the secret information, the PAN device can pair with other devices without again performing such a transmission.

[0025] In addition to further describing the concepts noted above, the following discussion describes how some embodiments utilize PAN devices to facilitate various wagering game features. The following discussion describes embodiments that use a PIN for pairing. Those embodiments are not limited to PINs, as they can use any suitable means for representing a shared secret.

Operating Environment

[0026] This section describes an example operating environment and presents structural aspects of some embodiments. This section includes discussion about wagering game machine architectures and wagering game networks.

Wagering Game Machine Architectures

[0027] FIG. **2** is a block diagram illustrating a wagering game machine architecture, according to example embodiments of the invention. As shown in FIG. **2**, the wagering game machine architecture **200** includes a wagering game machine **206**, which includes a central processing unit (CPU) **226** connected to main memory **228**. The CPU **226** can include any suitable processor, such as an Intel® Pentium processor, Intel® Core 2 Duo processor, AMD Opteron™ processor, or UltraSPARC processor.

[0028] The main memory **228** includes a PAN device controller **234**. The PAN device controller **234** can establish and control communications with PAN devices, such as PDAs, wireless headphones, remote controls, media players, etc. The PAN controller **234** works in concert with the PAN transceiver **236**, which can include logic for communicating according to any suitable PAN technology (e.g., Bluetooth™, IrDA®, ZigBee, Ultra-wideband, etc.). The PAN controller **234** and PAN transceiver **236** can detect and pair with PAN devices. In some instances, the PAN controller **234** requires a PIN for authenticating PAN devices. The PAN controller **234** can acquire the PIN via player input or from a central repository (e.g., an account server). The main memory **228** also includes a wagering game unit **232** that can present wagering games, such as video poker, video blackjack, video slots, video lottery, etc., in whole or part. The wagering game unit **232** can use the PAN components for transmitting content to PAN devices. In some instances, the wagering game unit **232** can present wagering games having functionality that requires PAN devices. For example, some wagering games can award MP3s or other content based on game results. For community games, the wagering game unit **232** can send player-specific output (e.g., audio) to a player's PAN device, so others cannot perceive the output.

[0029] The CPU **226** is also connected to an input/output (I/O) bus **222**, which can include any suitable bus technologies, such as an AGTL+frontside bus and a PCI backside bus. The I/O bus **222** is connected to a payout mechanism **208**, primary display **210**, secondary display **212**, value input device **214**, player input device **216**, information reader **218**, storage unit **230**, and the PAN transceiver **236**. The player input device **216** can include the value input device **214** to the extent the player input device **216** is used to place wagers. The I/O bus **222** is also connected to an

external system interface **224**, which is connected to external systems **204** (e.g., wagering game networks).

[0030] In one embodiment, the wagering game machine **206** can include additional peripheral devices and/or more than one of each component shown in FIG. **2**. For example, in one embodiment, the wagering game machine **206** can include multiple external system interfaces **224** and/or multiple CPUs **226**. In one embodiment, any of the components can be integrated or subdivided.

[0031] Any of the components of the architecture **200** can be integrated or subdivided. Also, any of the components can include hardware, firmware, and/or machine-readable media including instructions for performing the operations described herein. Machine-readable media includes any mechanism that provides (i.e., stores and/or transmits) information in a form readable by a machine (e.g., a wagering game machine, computer, etc.). For example, tangible machine-readable media includes read only memory (ROM), random access memory (RAM), magnetic disk storage media, optical storage media, flash memory machines, etc. Machine-readable media also includes any media suitable for transmitting software over a network.

[0032] While FIG. **2** describes an example wagering game machine architecture, this section continues with a discussion wagering game networks.

Wagering Game Networks

[0033] FIG. **3** is a block diagram illustrating a wagering game network **200**, according to example embodiments of the invention. As shown in FIG. **3**, the wagering game network **300** includes a plurality of casinos **312** connected to a communications network **314**. Each casino **312** includes a local area network **316**, which includes an access point **304**, a wagering game server **306**, account server **318**, PAN devices **320**, and wagering game machines **302**. The access point **304** provides wireless communication links **310** and wired communication links **308**. The wired and wireless communication links can employ any suitable connection technology, such as Bluetooth, 802.11, Ethernet, public switched telephone networks, SONET, etc. In some embodiments, the wagering game server **306** can serve single-player and community wagering games and distribute content to devices located in other casinos **312** or at other locations on the communications network **314**.

[0034] The PAN devices **320** can include any suitable devices, such as PDAs, wireless headsets, Wii™ Remotes from Nintendo®, wireless mice, wireless joysticks, etc. The PAN devices **320** can also include wired devices (e.g., wired Universal Serial Bus devices). As noted above, the PAN devices **320** can pair with wagering game machines **302** and exchange wagering game content and other information. In some instances, the wagering game machines **302** can act as Internet access points for the PAN devices **320**. That is, the wagering game machines **302** can access the Internet via the communications network **314** and exchange Internet content with the PAN devices **320** using PAN technologies.

[0035] During the pairing process, the wagering game machines **302** can receive PAN device PINs as player input or the machines **302** can procure PAN device PINs from the account server **318**. The account server **318** can store player account information including PAN device PINs, as described herein.

[0036] The wagering game machines **302** described herein can take any suitable form, such as floor standing models, handheld mobile units, bartop models, workstation-type console models, etc. Further, the wagering game machines **302** can be primarily dedicated for use in conducting wagering games, or can include non-dedicated devices, such as mobile phones, personal digital assistants, personal computers, etc. In one embodiment, the wagering game network **300** can include other network devices, such as accounting servers, wide area progressive servers, player tracking servers, and/or other devices suitable for use in connection with embodiments of the invention.

[0037] In some embodiments, wagering game machines **302** and wagering game servers **306** work together such that a wagering game machine **302** can be operated as a thin, thick, or intermediate

client. For example, one or more elements of game play may be controlled by the wagering game machine **302** (client) or the wagering game server **306** (server). Game play elements can include executable game code, lookup tables, configuration files, game outcome, audio or visual representations of the game, game assets or the like. In a thin-client example, the wagering game server **306** can perform functions such as determining game outcome or managing assets, while the wagering game machine **302** can present a graphical representation of such outcome or asset modification to the user (e.g., player). In a thick-client example, the wagering game machines **302** can determine game outcomes and communicate the outcomes to the wagering game server **306** for recording or managing a player's account.

[0038] In some embodiments, either the wagering game machines **302** (client) or the wagering game server **306** can provide functionality that is not directly related to game play. For example, account transactions and account rules may be managed centrally (e.g., by the wagering game server **306**) or locally (e.g., by the wagering game machine **302**). Other functionality not directly related to game play may include power management, presentation of advertising, software or firmware updates, system quality or security checks, etc.

[0039] Any of the wagering game network components (e.g., the wagering game machines **302**) can include hardware and machine-readable media including instructions for performing the operations described herein.

Example Operations

[0040] This section describes operations performed by some embodiments of the inventive subjective matter. In the discussion below, the flow diagrams will be described with reference to the block diagrams presented above. However, in some embodiments, the operations can be performed by logic not described in the block diagrams. In certain embodiments, the operations can be performed by executing instructions residing on machine-readable media (e.g., software), while in other embodiments, the operations can be performed by hardware and/or other logic (e.g., firmware). In some embodiments, the operations can be performed in series, while in other embodiments, one or more of the operations can be performed in parallel. Moreover, some embodiments can perform less than all the operations shown in any flow diagram.

Pairing PAN Devices

[0041] FIG. **4** is a flow diagram illustrating operations for pairing a wagering game machine with a PAN device, according to some embodiments of the invention. In FIG. **4**, the flow **400** begins at block **402**. At block **402**, a wagering game machine receives player credentials, such as by reading a player identifier from a player tracking card. The wagering game machine may require that the player enter a player password, too. The flow continues at block **404**.

[0042] At block **404**, the wagering game machine authenticates the player credentials. For example, the wagering game machine transmits the player credentials to an account server for comparison to stored versions of the credentials. In turn, the account server notifies the wagering game machine that the player credentials are authentic. The flow continues at block **406**.

[0043] At block **406**, the wagering game machine receives profile information associated with the player. The profile information includes one or more PAN device pins. In addition to the PAN device pins, the player profile information can include player preferences, funds, player game histories, etc. As described above, an account server or other repository can store pins for PAN devices that have previously paired with wagering game network devices (e.g., wagering game machines). The flow continues at block **408**.

[0044] At block **408**, the wagering game machine determines whether it detects a PAN device. If the wagering game machine detects a PAN device, the flow continues at block **410**. Otherwise, the flow continues at block **420**.

[0045] At block **410**, after detecting a PAN device, the wagering game machine determines whether the PAN device's PIN has been entered into the wagering game system. In making this determination, the wagering game machine can search the player's profile information for a PAN

device identifier (e.g., the PAN device's Media Access Control address or other hardware address) identical to that of the detected PAN device. For embodiments in which PAN device PINs are not part of the player's profile information, the wagering game machine can request a particular PAN device's PIN from a central repository or other peer devices. If the PAN device's PIN is not been stored in the wagering game system, the flow continues at block **412**. Otherwise, the flow continues at block **416**.

[0046] At block **412**, wagering game machine receives input indicating the PAN device's PIN. The input can be player input representing a PIN (or other secret information). Alternatively, the input can be from another device, such as the PAN device or a configuration device. The flow continues at block **414**.

[0047] At block **414**, the wagering game machine stores the PAN device's PIN. In some embodiments, wagering game machine transmits the PAN device PIN to the account server or other centralized repository for storage. Alternatively, the wagering game machine can store the PAN device PIN locally. The flow continues at block **416**.

[0048] At block **416**, the wagering game machine authenticates a PAN device using the PIN. In some embodiments, the machine authenticates the PAN device using asymmetric encryption based on the PIN. In other embodiments, the wagering game machine can authenticate the PAN device using techniques described in any of the Bluetooth™ specifications. For example, the wagering game machine and PAN device can use the PIN to create authentication keys, and to exchange messages created using one or more of the keys and other data (e.g., addresses, random numbers, etc.). If the machine and device verify the correctness of messages, authentication is successful. The flow continues at block **418**.

[0049] At block **418**, the wagering game machine establishes encrypted communications with the PAN device. The flow continues at block **420**.

[0050] At block **420**, the wagering game machine presents a wagering game using its own components and the PAN device. For example, if the PAN device is a wireless headset, the wagering game machine can transmit the wagering game's audio content to the headset for presentation. From block **420**, the flow ends.

[0051] While FIG. 4 describes pairing operations typically performed by wagering game machine, FIG. 5 describes pairing operations of a PAN device.

[0052] FIG. 5 is a flow diagram illustrating operations for pairing a PAN device to a wagering game machine, according to some embodiments of the invention. The flow **500** begins at block **502**. At block **502**, the PAN device receives a PIN. In some instances, a player can enter the PAN device's PIN via a keypad or other input device on the PAN device. In some PAN devices, the PAN device PIN is preset, so players need not enter the PIN (i.e., the PAN device does not perform the operation at block **402**). The flow continues at block **504**.

[0053] At block **504**, the PAN device uses the PIN (and possibly other data) to authenticate with a wagering game machine. In some embodiments, the PAN device and a wagering game machine will successfully authenticate if the PAN device's PIN (and possibly other information) matches a PIN received from the wagering game machine. In some embodiments, the PAN device can authenticate with a wagering game machine using techniques described in any of the Bluetooth™ specifications. The flow continues at block **506**.

[0054] At block **506**, the PAN device begins encrypted communications with the wagering game machine. In some embodiments, the encryption scheme is based on the PAN device PIN. The flow continues at block **508**.

[0055] At block **508**, the PAN device receives and presents content associated with a wagering game. In some embodiments, the PAN device receives audio or video content representing results of a wagering game, so a player can perceive the results on the PAN device instead of the wagering game machine. In some embodiments, the PAN device may receive media files or other content as part of an award for a wagering game presented on the wagering game machine. For example, the

PAN device may receive a link to a website at which the PAN device can select and download a free media file (e.g., MP3 file). From block **508**, the flow ends.

Using PAN Devices Without Player Cards

[0056] In some embodiments, PAN devices can perform functions typically performed by player tracking cards. That is, in some instances, PAN devices can electronically provide player credentials to wagering game machines, eliminating the need to insert player cards. In such embodiments, PAN devices can store player credentials, such as user identifiers and the like. After a PAN device has been “paired” with a wagering game machine, a player may end the gaming session, leaving with the device. When the player returns to another wagering game machine (or the same one), the PAN device can electronically provide player credentials (e.g., a user identifier) to the wagering game machine. After receiving the player credentials, the wagering game machine can present a screen showing the credential or a credential identifier, such as an avatar. In turn, the player can select his/her credential to confirm the transaction. The player may then need to enter a password (or other secret) at the wagering game machine. In some instances, the PAN device may transmit the password to the wagering game machine. In such an embodiment, a PAN device (e.g., a mobile phone) can display a message to “Press #1 to confirm login”. If the player presses #1 on the PAN device, the PAN device sends the player's password to the wagering game machine. In some instances, the PAN device may require separate player authentication before it will transmit the player's password.

[0057] To prevent a multitude of PAN devices from sending player credentials, the wagering game machine can filter out devices not within a specified proximity (e.g. via signal strength or locality mapping to other paired devices). Also, the wagering game machine may filter out (or place lower in the list) devices already paired with neighboring machines outside the wagering game machine's proximity.

[0058] In some embodiments, wagering game machines may repeatedly look for PAN devices that will provide player credentials. For example, if a wagering game machine having a zero credit balance suddenly receives money from a player, the machine may repeatedly look for a PAN device from which it can obtain player credentials. Similarly, if a player initiates a “sign-in” (e.g., by pressing a button on the wagering game machine), the machine will repeatedly look for a PAN device with which to pair. In some instances, a player may unexpectedly carry a PAN device out of range, causing the machine to terminate the player's game session. If a PAN moves out of range, the machine may repeatedly attempt to find the device. If the PAN remains out of range and there is value remaining on the wagering game machine (e.g., full or fractional credits), the machine can transfer the money back to the player's account or otherwise reallocate the value.

Community Wagering Games and PAN Devices

[0059] This section presents embodiments in which community wagering games direct presentation of certain results to PAN devices and other results to wagering game machine components. FIG. **6** describes these embodiments in more detail.

[0060] FIG. **6** is a flow diagram illustrating operations for using PAN devices to present community wagering game results, according to some embodiments of the invention. The flow **600** begins at block **602**. At block **602**, a wagering game system receives player inputs for a community wagering game. A plurality of wagering game machines can receive the player inputs and report them to a wagering game server or other device hosting the community game. The flow continues at block **604**.

[0061] At block **604**, a wagering game server or other host determines community results and player-specific results for the community wagering game. The community results depend on the type of community game. For example, for a community slots game, the community results can include a set of reel positions; for a community Texas Hold'Em game, the community results can include playing cards, etc. Player-specific results can include audible or visual hints that indicate how to increase a player's odds of winning in future community wagering games. For example, the

hint may reveal that certain game pieces (e.g., playing cards, slots reels, etc.) are more likely to appear. The player-specific results can also include other awards, such as music files, video files, photos, etc. The wagering game server can give awards based on player information (e.g., amounts wagered over a given time, time spent in a casino, etc.), casino information (e.g. a casino is offering special awards), or any other suitable information. The flow continues at block **606**.

[0062] At block **606**, the wagering game server determines whether PAN devices are available for players for whom player-specific results were determined. In some embodiments, the wagering game server can poll participating wagering game machines for a list of player identifiers and active PAN devices. Alternatively, participating wagering game machines can report this information before the community wagering game begins. If there are PAN devices available, the flow continues at block **608**. Otherwise, the flow continues at block **610**.

[0063] At block **608**, the wagering game server transmits the player-specific results for presentation on PAN devices. In some embodiments, the wagering game server can transmit content representing the player-specific results to wagering game machines paired with the relevant PAN devices. In turn, the wagering game machines can forward the results to the PAN devices. For example, if the player-specific results for a particular player include an audible hint, the wagering game server can transmit media content to a wagering game machine at which the player swiped a player card. In turn, the wagering game machine can forward the media content to a PAN device, which can reveal the hint by rendering the media content. In some instances, the wagering game server can select the content for representing the hint or other award based on the type of PAN device associated with the player. For example, if the PAN device is a headset, the hint will be included in an audio file. If the PAN device is a PDA, the hint can be included in a text file. Because a PAN device (e.g., a wireless headset, PDA, etc.) reveals the hint, it is more likely that only the player will perceive the hint. The flow continues at block **610**.

[0064] At block **610**, the wagering game server transmits the general results for presentation on the wagering game machines and other publicly perceptible devices (e.g., a video display specifically for the community game). Additionally, the server also transmits the general results addressed to any available PAN devices. As a result, players can view the general results on the wagering game machines and any available PAN devices. From block **610**, the flow ends.

[0065] While the discussion above indicates that wagering game machines (or other devices) transmit content to the PAN devices, so embodiments operate differently. Instead of transmitting the actual content, an identifier is transmitted, wherein the identifier identifies content already stored on the PAN device.

Additional Embodiments

[0066] In some embodiments, wagering game machines can use Sony's "TransferJet" technology to exchange content with PAN devices and other devices. The TransferJet technology is a close proximity wireless transfer technology enabling the high speed transfer of large data files (e.g., photos, HD images, etc.) between wagering game machines and mobile phones, digital cameras, digital video cameras, media players, etc. Using this technology, data can be sent at speeds of 560 Mbps.

[0067] Using TransferJet, just touching a wagering game machine with a device triggers an automatic download of media files. TransferJet eliminates complex setup procedures required by some existing wireless systems, and no access point is necessary. Players can register their electronic devices with a wagering game system so wagering game machines and other related devices recognize those devices. By registering devices, wagering game systems can prevent external data leakage.

[0068] In some embodiments in which a PAN device includes a Wii Remote™, the Wii Remote can be adapted to include a biometric reader, such as a vein scanner, fingerprint reader, retina scanner, etc. The Wii Remote's biometric reader can authenticate players. If a player's biometric information is authentic, the Wii Remote can provide the player's player identifier to a wagering game machine,

thus eliminating a need for the player card. If the player's biometric information is not authentic, the Wii Remote can remain locked.

Example Wagering Game Machines

[0069] This section provides more details about wagering game machines.

[0070] FIG. 7 is a perspective view of a wagering game machine, according to example embodiments of the invention. Referring to FIG. 7, a wagering game machine **700** is used in gaming establishments, such as casinos. According to embodiments, the wagering game machine **700** can be any type of wagering game machine and can have varying structures and methods of operation. For example, the wagering game machine **700** can be an electromechanical wagering game machine configured to play mechanical slots, or it can be an electronic wagering game machine configured to play video casino games, such as blackjack, slots, keno, poker, blackjack, roulette, etc.

[0071] The wagering game machine **700** comprises a housing **712** and includes input devices, including value input devices **718** and a player input device **724**. For output, the wagering game machine **700** includes a primary display **714** for displaying information about a basic wagering game. The primary display **714** can also display information about a bonus wagering game and a progressive wagering game. The wagering game machine **700** also includes a secondary display **716** for displaying wagering game events, wagering game outcomes, and/or signage information. While some components of the wagering game machine **700** are described herein, numerous other elements can exist and can be used in any number or combination to create varying forms of the wagering game machine **700**.

[0072] The value input devices **718** can take any suitable form and can be located on the front of the housing **712**. The value input devices **718** can receive currency and/or credits inserted by a player. The value input devices **718** can include coin acceptors for receiving coin currency and bill acceptors for receiving paper currency. Furthermore, the value input devices **718** can include ticket readers or barcode scanners for reading information stored on vouchers, cards, or other tangible portable storage devices. The vouchers or cards can authorize access to central accounts, which can transfer money to the wagering game machine **700**.

[0073] The player input device **724** comprises a plurality of push buttons on a button panel **726** for operating the wagering game machine **700**. In addition, or alternatively, the player input device **724** can comprise a touch screen **728** mounted over the primary display **714** and/or secondary display **716**.

[0074] The various components of the wagering game machine **700** can be connected directly to, or contained within, the housing **712**. Alternatively, some of the wagering game machine's components can be located outside of the housing **712**, while being communicatively coupled with the wagering game machine **700** using any suitable wired or wireless communication technology.

[0075] The operation of the basic wagering game can be displayed to the player on the primary display **714**. The primary display **714** can also display a bonus game associated with the basic wagering game. The primary display **714** can include a cathode ray tube (CRT), a high resolution liquid crystal display (LCD), a plasma display, light emitting diodes (LEDs), or any other type of display suitable for use in the wagering game machine **700**. Alternatively, the primary display **714** can include a number of mechanical reels to display the outcome. In FIG. 7, the wagering game machine **700** is an "upright" version in which the primary display **714** is oriented vertically relative to the player. Alternatively, the wagering game machine can be a "slant-top" version in which the primary display **714** is slanted at about a thirty-degree angle toward the player of the wagering game machine **700**. In yet another embodiment, the wagering game machine **700** can exhibit any suitable form factor, such as a free standing model, bartop model, mobile handheld model, or workstation console model.

[0076] A player begins playing a basic wagering game by making a wager via the value input device **718**. The player can initiate play by using the player input device's buttons or touch screen

728. The basic game can include arranging a plurality of symbols along a payline **732**, which indicates one or more outcomes of the basic game. Such outcomes can be randomly selected in response to player input. At least one of the outcomes, which can include any variation or combination of symbols, can trigger a bonus game.

[0077] In some embodiments, the wagering game machine **700** can also include an information reader **752**, which can include a card reader, ticket reader, bar code scanner, RFID transceiver, or computer readable storage medium interface. In some embodiments, the information reader **752** can be used to award complimentary services, restore game assets, track player habits, etc.

General

[0078] This detailed description refers to specific examples in the drawings and illustrations. These examples are described in sufficient detail to enable those skilled in the art to practice the inventive subject matter. These examples also serve to illustrate how the inventive subject matter can be applied to various purposes or embodiments. Other embodiments are included within the inventive subject matter, as logical, mechanical, electrical, and other changes can be made to the example embodiments described herein. Features of various embodiments described herein, however essential to the example embodiments in which they are incorporated, do not limit the inventive subject matter as a whole, and any reference to the invention, its elements, operation, and application are not limiting as a whole, but serve only to define these example embodiments. This detailed description does not, therefore, limit embodiments of the invention, which are defined only by the appended claims. Each of the embodiments is contemplated as falling within the inventive subject matter, which is set forth in the following claims.

Claims

1. A method comprising: authenticating, by a processor, a player account using, at least in part, a player identifier received at a first wagering game machine; detecting, by the processor a trigger condition indicating potential interaction with a wireless device associated with the authenticated player account; in response to detecting the trigger condition, automatically receiving, by the processor, without user input at the first wagering game machine, derived authentication data directly from the wireless device or from a configuration device accessible via a network via the communication interface, wherein the derived authentication data is based upon secret information including a wireless device identifier and a personal identification number (PIN) of the wireless device, the secret information having been previously stored in association with the player account information by an additional processor of a second wagering game machine communicatively connected to a network, in response to a previous pairing by the wireless device with the second wagering game machine; and automatically establishing, by the processor, a secure communication session between the first wagering game machine and the wireless device based, at least in part, on the received derived authentication data.
2. The method of claim 1, wherein the trigger condition comprises detecting a pairing initiation signal transmitted from the wireless device.
3. The method of claim 1, wherein the derived authentication data comprises at least one of a secure pairing credential derived from the secret information, a dynamic authentication token, or a previously established secure session identifier.
4. The method of claim 1, wherein automatically establishing the secure communication session comprises generating, by the processor based on the derived authentication data, an encryption key for the session.
5. The method of claim 1, wherein the trigger condition comprises detecting the wireless device in proximity while the player account is authenticated at the first wagering game machine.
6. The method of claim 1, wherein automatically establishing the secure communication session comprises validating, by the processor, the derived authentication data against session context

information.

7. The method of claim 1, wherein the derived authentication data is stored on the wireless device.

8. A wagering game machine comprising: a communication interface; a processor; and memory storing instructions that, when executed by the processor, cause the wagering game machine to: authenticate a player account using, at least in part, a player identifier received at the wagering game machine via the communication interface; detect a trigger condition indicating potential interaction with a wireless device associated with the authenticated player account; in response to detecting the trigger condition, automatically receive, without user input at the wagering game machine, derived authentication data directly from the wireless device or from a configuration device accessible via a network via the communication interface, wherein the derived authentication data is based upon secret information including a wireless device identifier and a personal identification number (PIN) of the wireless device, the secret information having been previously stored in association with the player account information by an additional processor of an additional wagering game machine communicatively connected to a network, in response to a previous pairing by the wireless device with the additional wagering game machine; and automatically establish a secure communication session with the wireless device based, at least in part, on the received derived authentication data.

9. The wagering game machine of claim 8, wherein the trigger condition comprises the communication interface detecting a pairing initiation signal transmitted from the wireless device.

10. The wagering game machine of claim 8, wherein the derived authentication data comprises at least one of a secure pairing credential derived from the secret information, a dynamic authentication token, or a previously established secure session identifier.

11. The wagering game machine of claim 8, wherein the instructions to automatically establish the secure communication session comprise instructions to validate the derived authentication data against session context information stored in the memory.

12. The wagering game machine of claim 8, wherein the instructions to automatically establish the secure communication session comprise instructions to generate, based on the derived authentication data, an encryption key for the session.

13. The wagering game machine of claim 8, wherein the trigger condition comprises detection of the wireless device in proximity while the player account is authenticated at the wagering game machine.

14. The wagering game machine of claim 8, wherein the derived authentication data is stored on the wireless device.

15. One or more non-transitory machine-readable storage mediums storing instructions that, when executed by a processor of a wagering game machine in a wagering game network, cause the wagering game machine to perform operations comprising: receiving a player identifier associated with an authenticated player account; in response to receiving the player identifier, determining whether pairing information associated with the player identifier for a specific wireless device is stored on a centralized storage device accessible via a communication network, wherein the pairing information was previously stored on the centralized storage device based on a previous pairing or registration involving the specific wireless device, and wherein the pairing information includes one or more security elements based upon the specific wireless device; in response to determining that the pairing information is stored on the centralized storage device, retrieving, via the communication network, the stored pairing information from the centralized storage device; and automatically establishing a secure communication session with the specific wireless device based, at least in part, on the retrieved pairing information, wherein said automatically establishing the secure communication session includes performing a cryptographic authentication process with the specific wireless device utilizing the retrieved pairing information.

16. The one or more non-transitory machine-readable storage mediums of claim 15, wherein the one or more security elements comprise at least one of a personal identification number (PIN)

associated with the specific wireless device or a unique wireless device identifier for the specific wireless device.

17. The one or more non-transitory machine-readable storage mediums of claim 15, wherein the one or more security elements comprise both a personal identification number (PIN) associated with the specific wireless device and a unique wireless device identifier for the specific wireless device.

18. The one or more non-transitory machine-readable storage mediums of claim 15, wherein the one or more security elements comprise at least one of an authentication code including suitable indicia associated with the specific wireless device or a cryptographic key associated with the specific wireless device.

19. The one or more non-transitory machine-readable storage mediums of claim 15, wherein the one or more security elements comprise data referencing a biometric template associated with an authorized user of the specific wireless device, wherein the biometric template was captured and stored during the previous pairing or registration involving the specific wireless device.
